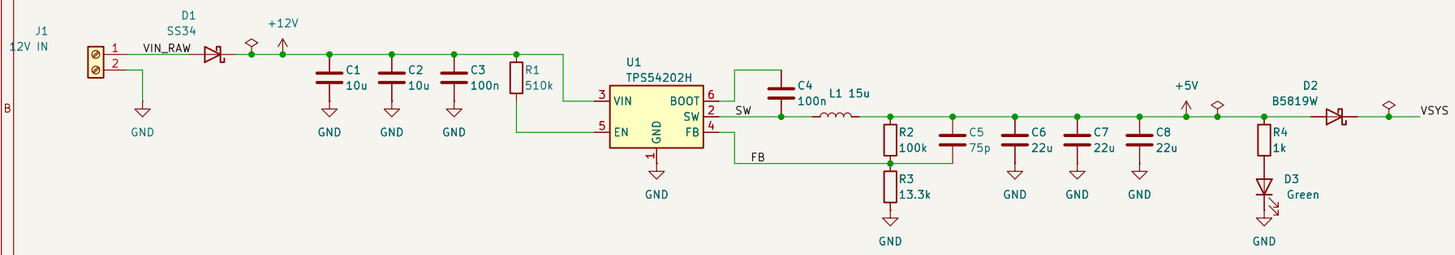
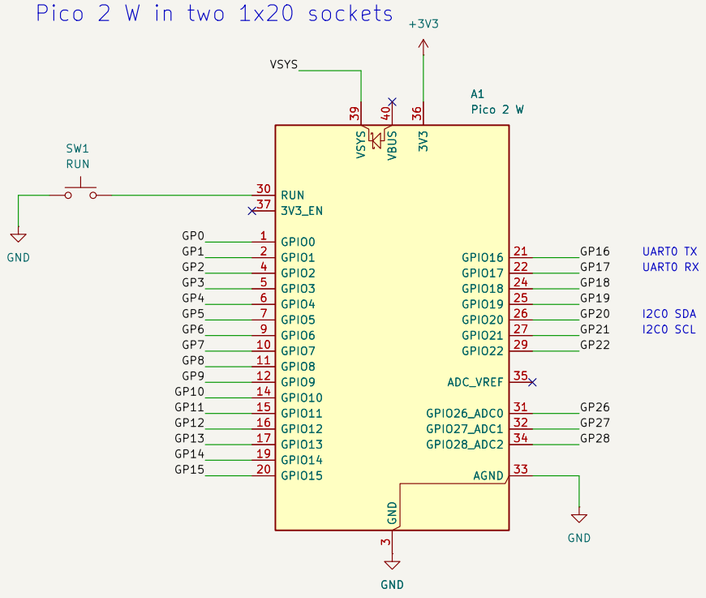
12 V in, 5 V buck



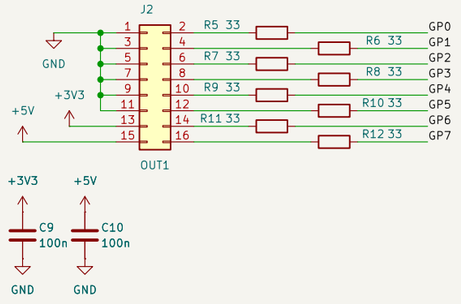
U1 and its parts are the led-processor-board buck (TI's TPS54202HEVM-716 5 V design): 100k / 13.3k with 75 pF across the top resistor, 0.596 V x (1 + 100/13.3) = 5.08 V; 15 uH; 500 kHz. The 510k enables U1 from the 12 V side. D1 blocks a reversed 12 V lead.
+5V feeds ribbon pin 15 directly. The Pico gets it on VSYS through D2 (about 4.7 V), so USB and 12 V can both be plugged in: the Pico's own diode keeps VSYS off the USB port and D2 keeps USB out of the buck.
On USB alone the Pico runs but the ribbons have no +5V, as on the processor board.
The load is small: the Pico and two output boards' level shifters, well under 0.5 A.

Pico 2 W in two 1x20 sockets

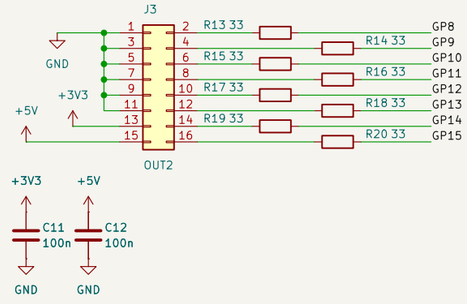


GP0-7 = OUT1, GP8-15 = OUT2: two runs of eight consecutive pins, one PIO state machine each in the PicoLEDs renderer.
GP16/17 = debug UART (UART0), GP20/21 = I2C0.
GP23-25 and GP29 are the Pico's own (radio, LED, VSYS sense).
USB, BOOTSEL, SWD and the LED are on the Pico. SW1 pulls RUN low to reset it.

OUT1: GP0-7

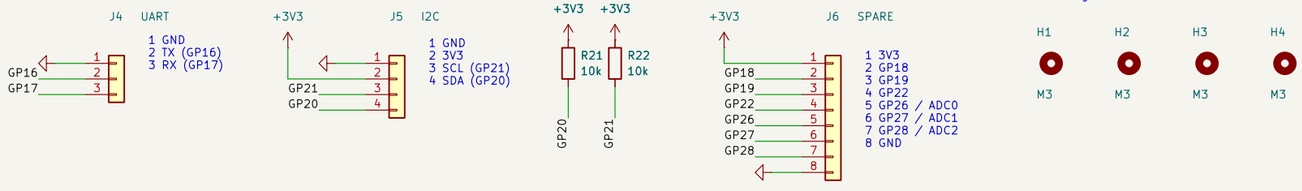


OUT2: GP8-15



Output shrouds: 16-pin IDC to the output boards. Pin 1 is the triangle on the shroud.
Odd pins 1-11 GND, 13 = +3V3 (shifter VCCA), 15 = +5V (shifter VCCB); data 1-8 on even pins 2-16 (same as led-output-board J2 and led-processor-board). 33R series resistors at the Pico end.

Headers



J4: debug UART, moved off GP0/1 so the outputs get GP0-15.
J5: I2C0 with 10k pull-ups, for a small display or a sensor.

Power and outputs follow led-processor-board

Sheet: /
File: led-budget-board.kicad_sch

Title: LED budget board (Pico 2 W carrier, 16 strands)

Size: A3	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1